

Curtain UI: Augmenting Curtains for Tangible Interactions

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Abstract

Curtains are functional textiles that play a crucial yet invisible role in our daily lives. Despite their widespread use for decoration, blocking heat, privacy, etc., they are still passive in our surroundings. This paper presents a design exploration and development of an interactive curtain interface using capacitive sensing. We augment the capabilities of everyday curtains into touch-sensitive surfaces to facilitate embodied interactions through physical manipulation and gestures. By interfacing these curtains with a smart home environment to control lights, fans, and other appliances, we show that interactive curtains are technically feasible, thus paving the way for novel Curtain UIs as a medium of tangible, embodied, and embedded interaction in a ubiquitous computing scenario.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Interaction design.

Keywords

Interactive Curtain, Curtain Based Interface

ACM Reference Format:

Pranshu Anand, Vishal Bharti, and Anmol Srivastava. 2024. Curtain UI: Augmenting Curtains for Tangible Interactions. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697935>

1 INTRODUCTION

Ubiquitous computing has emerged as a promising field in human-computer interaction, offering a unique approach to seamlessly integrating technology into our everyday lives; as predicted by Mark Weiser, profound technology will blend into the fabric, making them one unit [Weiser, 1999]. While published studies have explored the potential of spaces [Funk et al., 2015; Olwal et al., 2018; Zhang et al., 2018] as a means of user interactions, curtains have often been overlooked and present a potential opportunity for enabling embedded and embodied interactions. We propose and develop a novel Curtain UI, i.e., a curtain-based user interface capable of sensing touch and manipulation, using a Research through Design (RtD) approach [Zimmerman et al., 2007]. This curtain interface was integrated with a smart home environment, allowing it to control various appliances. Our research presents novel spatial

experiences that can be generated by harnessing the curtain's physical characteristics by leveraging ubiquitous computing principles and embodied interaction [Dourish, 2001].

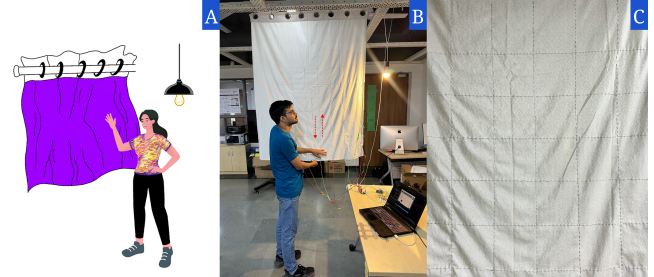


Figure 1: (A) Scenario of Curtain UI for controlling IoT-enabled devices (B) User activating a bulb via Curtain UI, (C) Curtain UI made with conductive thread

2 RELATED WORK

Ubiquitous computing has recently been applied to a significant number of everyday objects, making spaces and artefacts engaging media. Curtains represent an overlooked opportunity for innovation in this space.

Although Takashina et al., 2015 and Funk et al., 2015 explored curtains as interactive displays and media interfaces, their work primarily viewed curtains as display surfaces and did not delve into the potential for curtain-based interaction. Wall++ [Zhang et al., 2018] treated ordinary walls as smart infrastructures, augmenting rooms sensing and interactivity beyond conventional divider roles of walls.

The integration of sensors into fabric for interactivity is currently an area of important research. Early work by Post and Orth, 1997 and Orth et al., 1998 demonstrated smart fabrics with integrated capacitive sensing. Later developments on this include Poupyrev et al., 2016 Project Jacquard which further showed the potential of fabric-based interactions. Schoessler et al., 2015 used cords as input devices in Cord UI, while Olwal et al., 2018 developed I/O Braid: an interactive textile cord with embedded sensing and visual feedback. Building on these studies' insights and filling the existing gaps in research with regard to curtain-based interaction, we present Curtain UI to uncover curtains' interactive potential within the paradigm of ubiquitous computing.

3 PROTOTYPE DEVELOPMENT

The Curtain UI (Figure 1) project adopted the process of a Research through Design approach that tried to make a curtain interactive

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SA Posters '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1138-1/24/12

<https://doi.org/10.1145/3681756.3697935>

with capacitive sensing in 2d Matrix. The research was accomplished, with different materials applied on the curtain in a direct way to optimize its functionality and user interaction.

3.1 Material Exploration and Development

The project began by integrating aluminum tapes (Figure 2A) on curtain fabric, configured in a 6x5 grid and connected to a Bare Conductive [Bare Conductive, 2024]. This initial setup was designed to detect changes in capacitance caused by touch, utilizing Processing software [Processing Foundation, 2024] for real-time feedback, recognition of curtain surface manipulations, and attempts at detecting complex gestures.

The exploration progressed with the introduction of copper fabric tape (Figure 2B), chosen for its enhanced flexibility and durability. Configured in a 6x6 grid on the curtain significantly improved the system's ability to detect gestures such as swipes and taps. Advanced noise reduction techniques further refined the gesture recognition capabilities, facilitating practical integration with an electric bulb through relay-controlled appliances.

In the final phase of exploration, conductive thread (Figure 2C) was stitched into the curtain's fabric in a 6x6 grid, offering enhanced flexibility and better integration with the fabric. This material outperformed earlier ones by making the curtain feel more natural and less obstructive, significantly increasing the accuracy of gesture detection. Unlike aluminum and copper fabric tapes, which could become brittle and degrade over time due to physical interactions, conductive thread maintained its functional integrity. This allowed for more precise and reliable recognition of gestures, crucial for distinguishing between different swipe directions and thereby enhancing usability. The prototype was connected to an electric bulb via a relay controlled by a Bare Conductive Touch Board [Bare Conductive, 2024]. Swiping up on the curtain turned the bulb on, and swiping down turned it off. Additionally, the system could accurately detect swipe right, swipe left, and various tap patterns, enabling connections to other smart home functionalities and enhancing the interactive potential of the Curtain UI.

3.2 Comparative Evaluation

All phases of the material exploration were very informative in a useful sense and instrumental in the curtain's transition into an interactive interface. From aluminum tape, foundational insights could be gained; shifting to copper fabric tape improved it in flexibility and durability. Then, conductive thread increased the performance greatly with quite a heightened accuracy and reliability compared to gesture detection.

4 FUTURE WORK

Future improvements will aim to make these prototypes more robust and scalable. This will involve integrating machine learning algorithms for more complex gesture recognition, and enabling user customization to personalize interactions within the smart home environment. These enhancements will further extend the interactive potential of curtains.

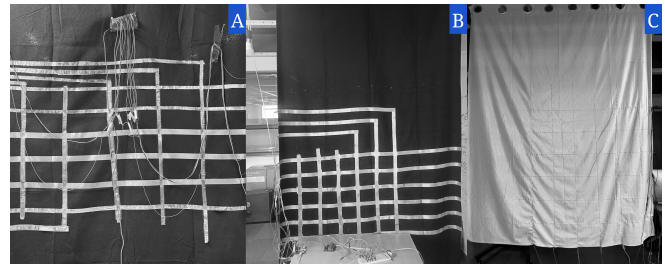


Figure 2: Curtain UI Explorations (A) Aluminum Tape (B) Copper Fabric Tape (C) Conductive Thread

Acknowledgments

Pranshu Anand conducted this research at the Creative Interfaces Lab as part of his Master's coursework and was the recipient of the Chanakya Fellowship, supported by iHub Anubhuti - IIITD Foundation. This work was also supported by the Centre for Design and New Media, a TCS Foundation Initiative at IIIT Delhi. We sincerely thank the fellow researchers at the Creative Interfaces Lab for their unwavering support throughout this project.

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